

The Role of AI

Algorithm to bedside symposia





DATA STORIES

From algorithm to bedside

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Senior Forsker

Gentofte or Herlev Hospital

03-14 Feb. 2025

Lunch symposia

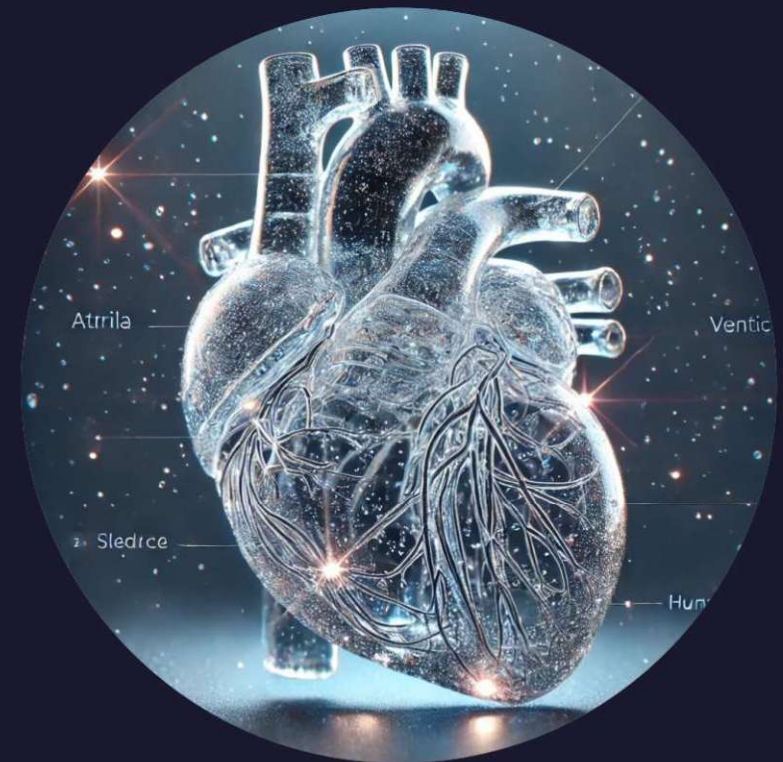
Omics-Based Clinical and Population Studies (**session 1: Mon. 3 Feb. 2025** - Specific Case Studies)

Contextual Epidemiology of Cardiometabolic Conditions, Polypharmacy and Multi-morbidity (**session 2: Thurs. 6 Feb. 2025**)

Scientific Methods in Biomedical Research (**sessions 3 and 4: Mon and Thurs. 10 and 13 Feb. 2025**)

- Omics in biomedical research
- Evidence pyramid and analytics
- Global Burden of Disease, Health Analytics, and Access to Therapeutics
- Economic Evaluations

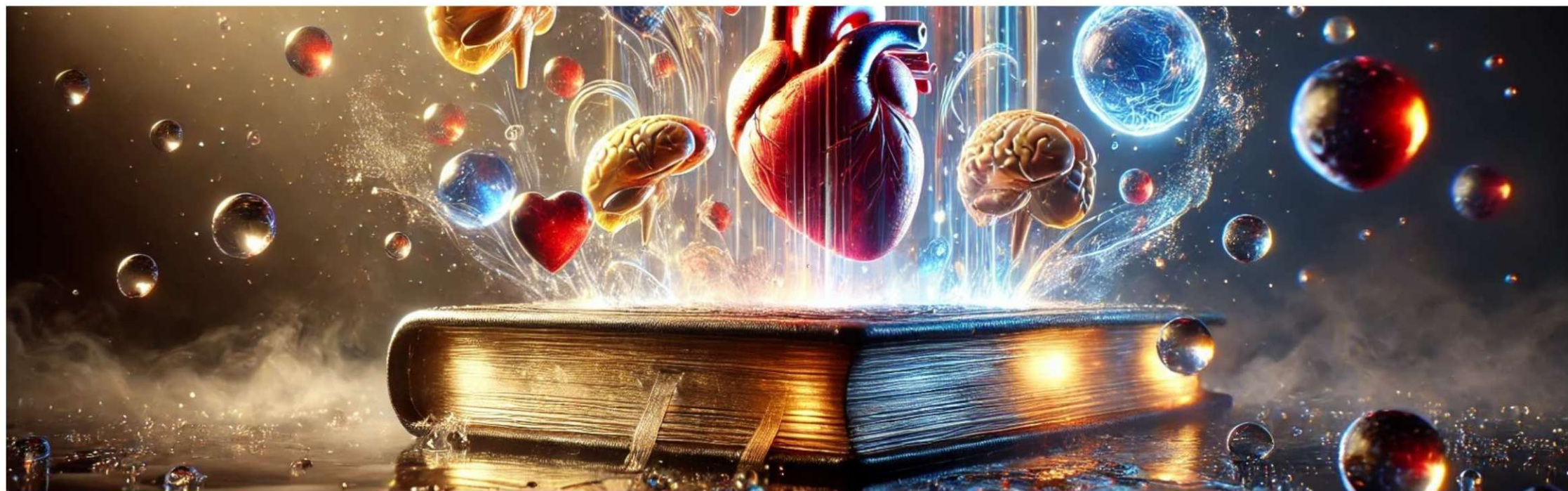
Github link: <https://github.com/Nermin-Ghith/From-algorithm-to-bedside>





AI AND MIRCO to Macro LEVEL **HEALTH** A NEW FRONTIER





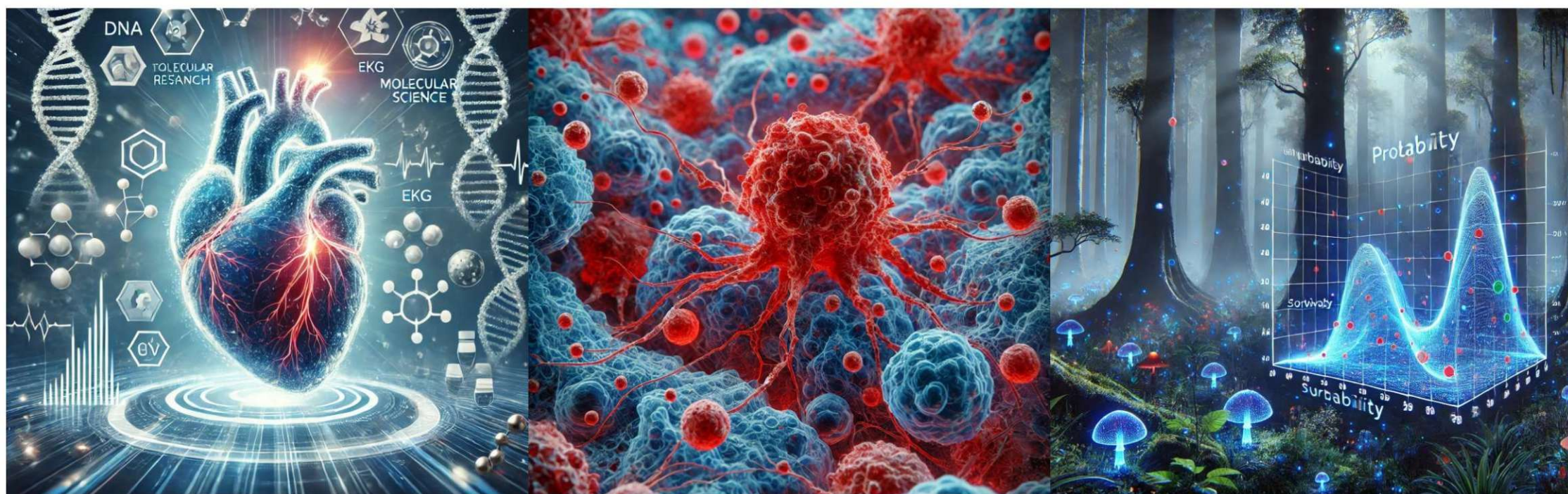
Session 4

- Omics in biomedical research - Clinical trials and risk scores
- Evidence pyramid and analytics - Clinical trials and risk scores
- Global Burden of Disease, Health Analytics, and Access to Therapeutics
- Economic Evaluations

From algorithm to bedside

Risk Scores





RSs in clinical research

- RCTS
- Clinical practice
- (Metagenomic) population studies
- Quality of patient care and survival
- Polygenic RSs in prevention

Patient risk scores (RS)

- This is the **RS** for developing critical illness or health outcome (e.g., mortality, readmission, negative prognosis).
- Applied to a **patient** or a population
- Population risk scores:
 - For **screening** purposes.
 - In primary care, with the help of software.
- With any **design**
- Assign the proper level of **severity** and **complexity** of the illness



RS in RCTs

Eligibility for treatments

Estimate the risk of death among critically ill patients

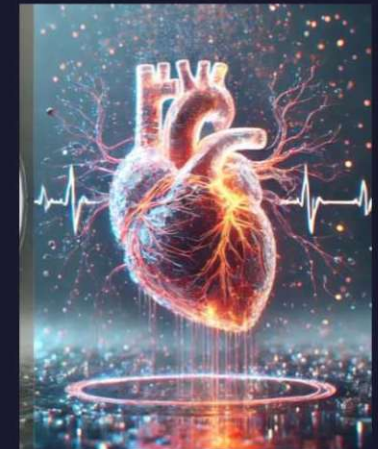
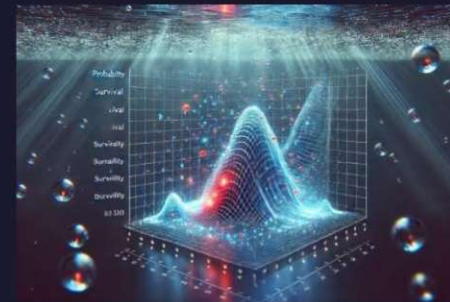
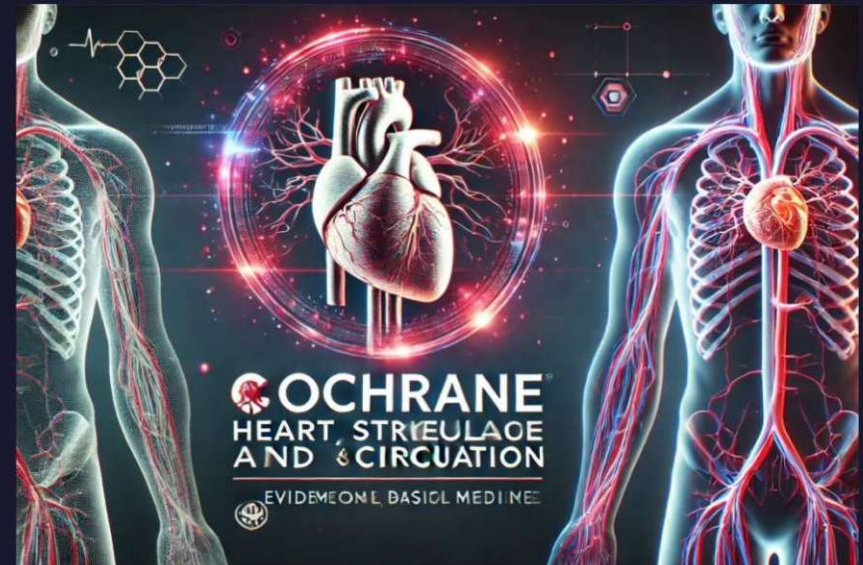
APACHE II scores: double-blind, placebo-controlled, multicenter trial, to assess the efficacy of recombinant human activated protein C, or drotrecogin alfa (activated) (DrotAA), for the treatment of patients with septic shock. (DOI: 10.1056/NEJMoa1202290).

Risk stratification (severity of illness)

- APACHE II scores: Placebo-controlled, multicenter RCT to assess the Efficacy and Safety of Recombinant Human Activated Protein C for Severe Sepsis. (DOI: 10.1056/NEJM200103083441001)
 - APACHE II: the score on the Acute Physiology and Chronic Health Evaluation II.

In subgroup analyses:

- **Framingham risk score:** RCT to assess Rosuvastatin (20 mg daily or placebo) use to Prevent Vascular Events in Men and Women with Elevated C-reactive protein. (<https://doi.org/10.1056/nejmoa0807646>).



Framingham Heart Study



FRAMINGHAM RISK SCORE (FRS) Estimation of 10-year Cardiovascular Disease (CVD) Risk

Step 1¹
In the "points" column enter the appropriate value according to the patient's age, HDL-C, total cholesterol, systolic blood pressure, and if they smoke or have diabetes. Calculate the total points.

Risk Factor	Risk Points		Points	
	Men	Women		
Age				
30-34	0	0		
35-39	2	2		
40-44	5	4		
45-49	7	5		
50-54	8	7		
55-59	10	8		
60-64	11	9		
65-69	12	10		
70-74	14	11		
75+	15	12		
HDL-C (mmol/L)				
>1.6	-2	-2		
1.3-1.6	-1	-1		
1.2-1.29	0	0		
0.9-1.19	1	1		
<0.9	2	2		
Total Cholesterol				
<4.1	0	0		
4.1-5.19	1	1		
5.2-6.19	2	3		
6.2-7.2	3	4		
>7.2	4	5		
Systolic Blood Pressure (mmHg)	Not Treated	Treated	Not Treated	Treated
<120	-2	0	-3	-1
120-129	0	2	0	2
130-139	1	3	1	3
140-149	2	4	2	5
150-159	2	4	4	6
160+	3	5	5	7
Smoker	Yes	4	No	3
No	0	0	0	0
Diabetes	Yes	statin-indicated condition		
No	0		0	
Total Points				

Step 2²
Using the total points from Step 1, determine the 10-year CVD risk* (%).

Total Points	10-Year CVD Risk (%)	
	Men	Women
<3 or less	<1	<1
-2	1.1	1.0
-1	1.4	1.2
0	1.6	1.5
1	1.9	1.7
2	2.3	2.0
3	2.8	2.4
4	3.3	2.8
5	3.9	3.3
6	4.7	3.9
7	5.6	4.5
8	6.7	5.3
9	7.9	6.3
10	9.4	7.3
11	11.2	8.6
12	13.3	10.0
13	15.6	11.7
14	18.4	13.7
15	21.6	15.9
16	25.3	18.5
17	29.4	21.5
18	30	24.6
19	30	27.5
20	30	30
21+	30	30

Step 3³
Using the total points from Step 1, determine heart age (in years).

Heart Age, y	Men	Women
<30	<0	<1
30	0	1
31	1	2
32	2	3
33	3	4
34	4	5
35	5	6
36	6	7
37	7	8
38	8	9
39	9	10
40	10	11
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317	287	28

Framingham Heart Study

Risk Functions



Risk Functions →

Atrial Fibrillation 10 year Risk

Heart Failure in Atrial Fibrillation (10-year risk)

Cardiovascular Disease (10-year risk)

Cardiovascular Disease (30-year risk)

Congestive Heart Failure

Hard Coronary Heart Disease (10-year risk)

Coronary Heart Disease (10-year risk)

Riesgo de Enfermedades Coronarias Cardiacas en 10 Años

Recurrent Coronary Heart Disease

Coronary Heart Disease (2-year risk) – First Event

Diabetes

Fatty Liver Disease Risk Function

Hypertension

Intermittent Claudication

Stroke

Type 2 diabetes-specific risk calculator

UKPDS Risk Engine v2.0

Input

Age Now : 62 years HbA1c : 8.3 %
Duration of Diabetes : 11 years Systolic BP : 145 mmHg
Sex : ☒ Male ☐ Female Total Cholesterol : 5.8 mmol/l
Atrial Fibrillation : ☒ No ☐ Yes HDL Cholesterol : 1.1 mmol/l
Ethnicity : White
Smoking : Non-Smoker

Options >

Output

10 year risk 0 15 30 100

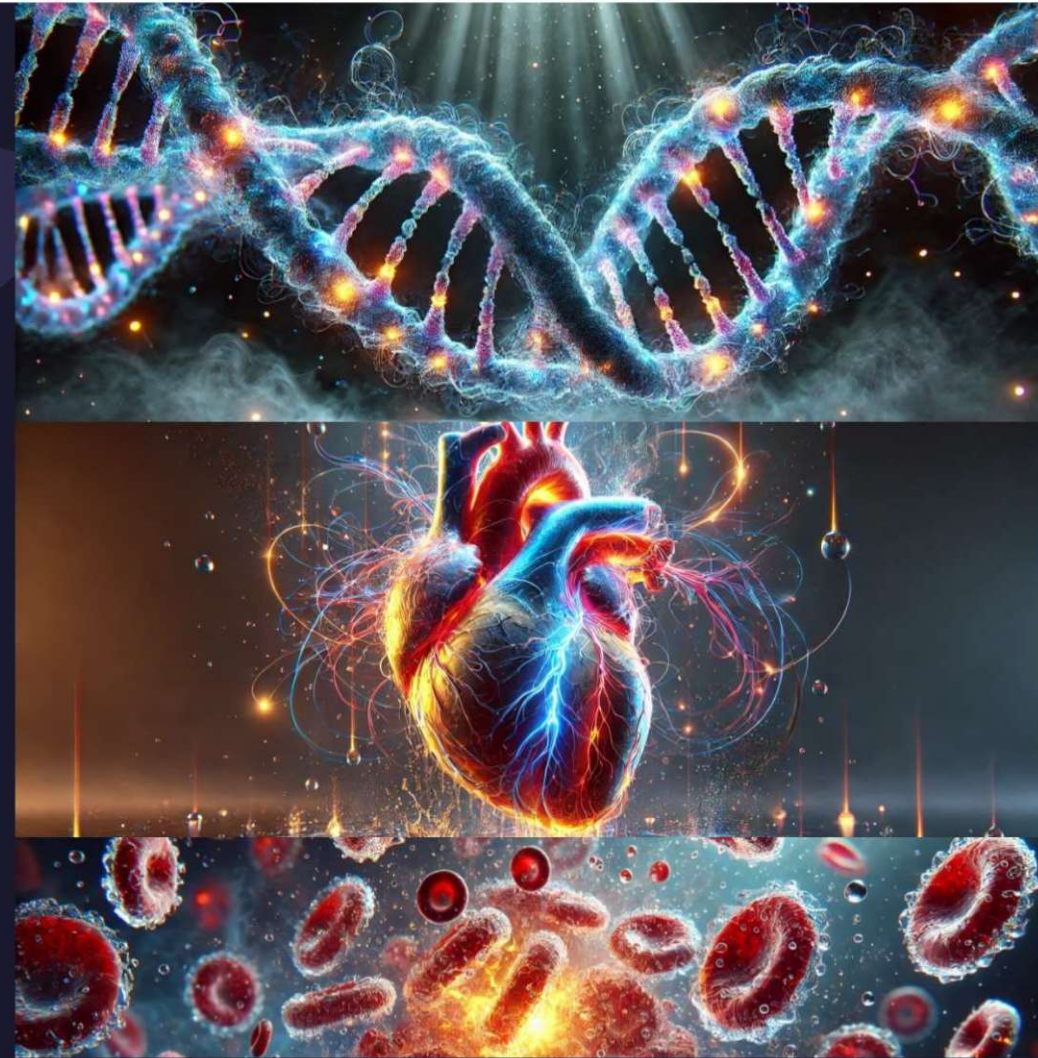
CHD :	33.3%	
Fatal CHD :	24.4%	
Stroke :	11.6%	
Fatal Stroke :	1.8%	

Adjusted for regression dilution

Details Copy Print
Help Exit

RS in clinical practice/studies

- Charlson Comorbidity Index
- Johns Hopkins Aggregated Diagnosis Groups (ADGs) - Predict Mortality in a General Adult Population
- Elixhauser Comorbidity Score - to predict 30-day, in-hospital, and 1-year mortality in older adults
- QRISK3 risk prediction algorithms to estimate future risk of cardiovascular disease
- The HAS-BLED score – risk of bleeding in patients with atrial fibrillation.

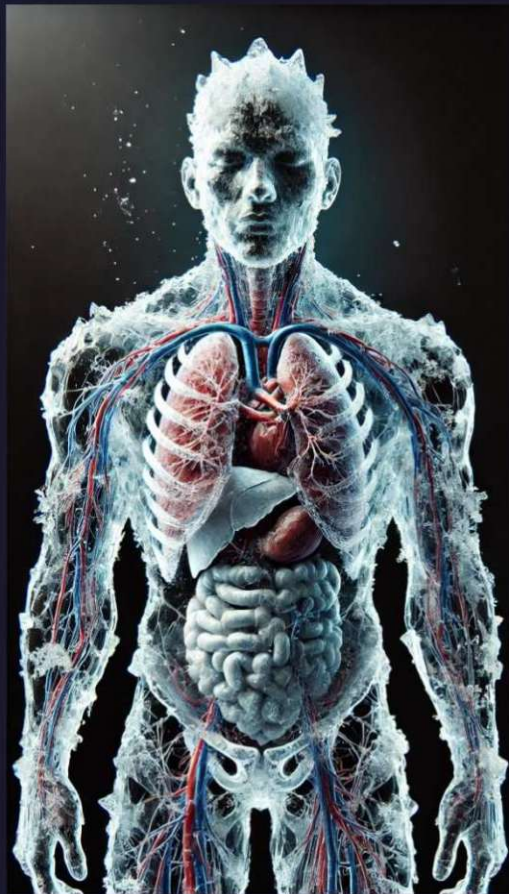


Charlson Comorbidity Index



- **Charlson Comorbidity Index (CCI), 1987**, is an assessment tool used to predict **long-term mortality** across various **clinical** populations (e.g., medical, surgical, ICU, trauma, and cancer patients).
- It can also predict **in-hospital mortality**, though it may not perform as well for certain groups like **ICU** or trauma patients when compared to tools designed specifically for those situations.
- The CCI is useful in providing a **valid assessment** of a patient's unique clinical condition.
- It helps **differentiate** major diagnostic and prognostic differences among patients with the same medical diagnosis.

- **weighted scoring system** - predict the **risk of death within one year** after hospitalization for patients with certain **comorbid/multi conditions**.
- The index includes **19 conditions**, each assigned a **weight from 1 to 6** based on its impact on **1-year mortality**



Conditions	Assigned weights for each condition
MI	1
CHF	1
Peripheral vascular disease	1
Cerebrovascular disease	1
Dementia	1
Chronic pulmonary disease	1
Connective tissue disease	1
Ulcer disease	1
Mild liver disease	1
Diabetes	1
Hemiplegia	2
Moderate or severe renal disease	2
Diabetes with end organ damage	2
Any tumor without metastasis	2
Leukemia	2
Lymphoma	2
Moderate or severe liver disease	3
Metastatic solid tumor	6
AIDS	6

MI, myocardial infarction; CHF, congestive heart failure.



Re-weighted versions of the CCI

Conditions	Original CCI	Quan's "Updated Charlson"	Ghali CABG	Klabunde prostate	Klabunde breast	Chaundhury inpatients	Schneeweiss elderly	Elixhauser hospitalized patients
MI	1		1	1	3		1	
Peripheral vascular disease	1		2	1	1		1	1
Cerebrovascular disease	1		1	1			1	
CHF	1	2	4	2	2		1	1
Peptic ulcer disease	1							1
Diabetes	1			1	2		1	1
Chronic pulmonary disease	1	1		2	2		2	1
Connective tissue disease	1	1		3	3			1
Dementia	1	2					3	
Mild liver disease	1	2		2	3		2	1
Diabetes with end organ damage	2	1		2	2		2	1
Moderate or severe renal disease	2	4	3	2	4	2	3	1
Hemiplegia	2	2		1	3		1	
Any tumor without metastasis	2	2				2	2	1
Leukemia	2	2					2	
Lymphoma	2	2					2	1
Moderate or severe liver disease	3	4				2	3	
AIDS	6	4				3	6	1
Metastatic solid tumor	6	6					6	1
<i>Additional Elixhauser conditions</i>								
Arrhythmia								1
Coagulation								1
Fluid electrolyte								1
Alcohol dependence								1
Pulmonary circulation disorder								1
Hypertension								1
Hypothyroidism								1
Obesity								1
Weight loss								1
Anemia (blood loss)								1
Deficiency anemia								1
Drug abuse								1
Psychosis								1
Depression								1
Neurologic or degenerative disorder								1
Paralysis								1
Citations	39,225	2,849	296	1,644	1,644	309	1,569	7,361

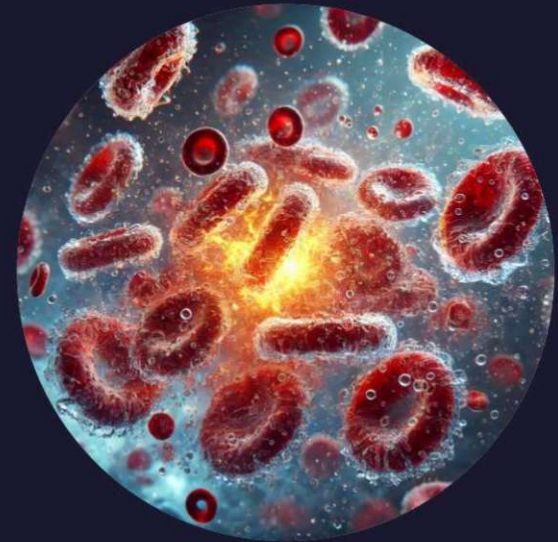
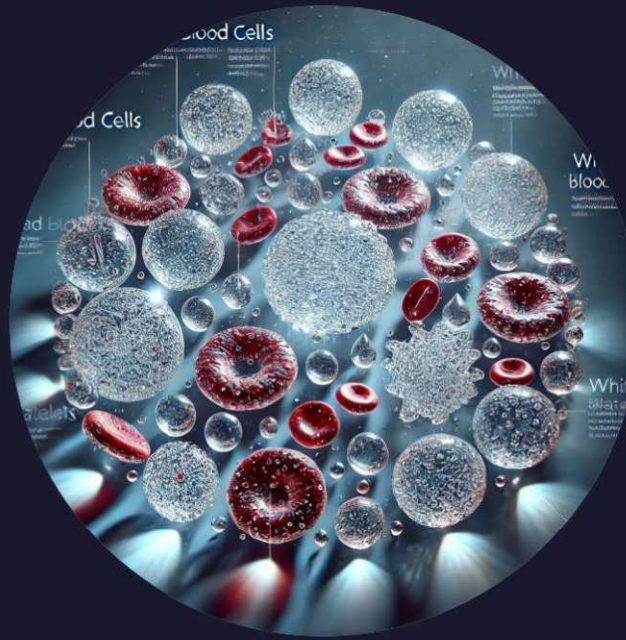
MI, myocardial infarction; CHF, congestive heart failure.



HAS-BLED score

Bleeding Risk in Anticoagulated Patients With Atrial Fibrillation

- The HAS-BLED score, 2010, is a real-world cohort of 3450 anticoagulated patients with AF.
- (hypertension, abnormal renal/liver function, stroke, bleeding history or predisposition, labile international normalized ratio [INR], elderly [age ≥ 65 years], drugs/alcohol concomitantly)
- 3 risk stratifications: a score of 0 indicates low risk, 1–2 indicates moderate risk, and ≥ 3 indicates high risk. Recently.
- **Overlapping risk scores: HEMORR₂HAGES, ATRIA, CHADS₂, or CHA₂DS₂-VASc scores**



RESEARCH ARTICLE

Short Term Survival after Admission for Heart Failure in Sweden: Applying Multilevel Analyses of Discriminatory Accuracy to Evaluate Institutional Performance

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OPEN ACCESS

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Data Availability Statement: The database we analyzed is not publicly available for ethical and data safety reasons according to the Swedish National Board of Health and Welfare. However, the same dataset can be constructed by request to the Swedish National Board of Health and Welfare after approval of the research project by an Ethical Committee and by the data safety committee at the Swedish National Board of Health and Welfare.

Funding: This work was supported by the Swedish Research Council (VR) [82013-2484, Juan Merlo] and by grants from the University of Copenhagen and

Abstract

Background

Hospital performance is frequently evaluated by analyzing differences between hospital averages in some quality indicators. The results are often expressed as quality charts of hospital variance (e.g., league tables, funnel plots). However, those analyses seldom consider patients heterogeneity around averages, which is of fundamental relevance for a correct evaluation. Therefore, we apply an innovative methodology based on measures of components of variance and discriminatory accuracy to analyze 30-day mortality after hospital discharge with a diagnosis of Heart Failure (HF) in Sweden.

Methods

We analyzed 36,943 patients aged 45–80 treated in 565 wards at 71 hospitals during 2007–2009. We applied single and multilevel logistic regression analyses to calculate the odds ratios and the area under the receiver-operating characteristic (AUC). We evaluated general hospital and ward effects by quantifying the intra-class correlation coefficient (ICC) and the increment in the AUC obtained by adding random effects in a multilevel regression analysis (MLRA). Finally, the Odds Ratios (ORs) for specific ward and hospital characteristics were interpreted jointly with the proportional change in variance (PCV) and the proportion of ORs in the opposite direction (POOR).

Findings

Overall, the average 30-day mortality was 9%. Using only patient information on age and previous hospitalizations for different diseases we obtained an AUC = 0.727. This value was almost unchanged when adding sex, country of birth as well as hospitals and wards levels. Average mortality was higher in small wards and municipal hospitals but the POOR values were 15% and 16% respectively.

RS - quality of patient care and survival



<https://doi.org/10.1371/journal.pone.0148187>





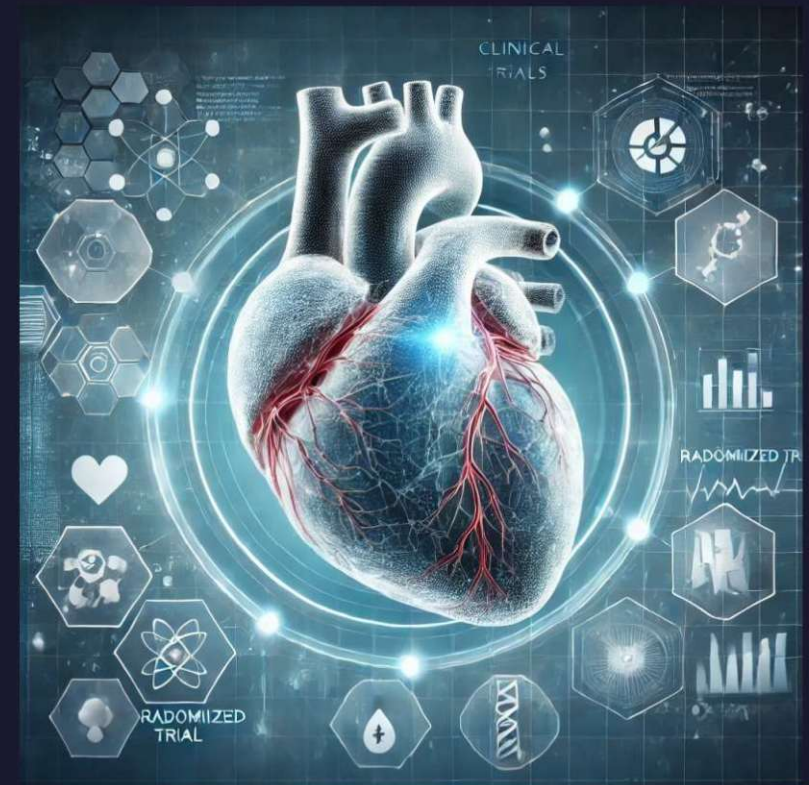
RS - quality of patient care and survival

- Study (patient population) specific risk score – Ghith et al. (2016):
<https://doi.org/10.1371/journal.pone.0148187>
- knowing the patient age and previous diseases (i.e., **the RS**) was enough to achieve an AUC = 0.727 which was only slightly better when knowing the sex and ethnic status of the patient (+0.002 units).
- However, this patient-level information was only marginally improved by knowing the **hospital or the ward** where the patient was treated.

Study I – Methods and findings

Data: using a unique personal identification number, the research database included linked data from

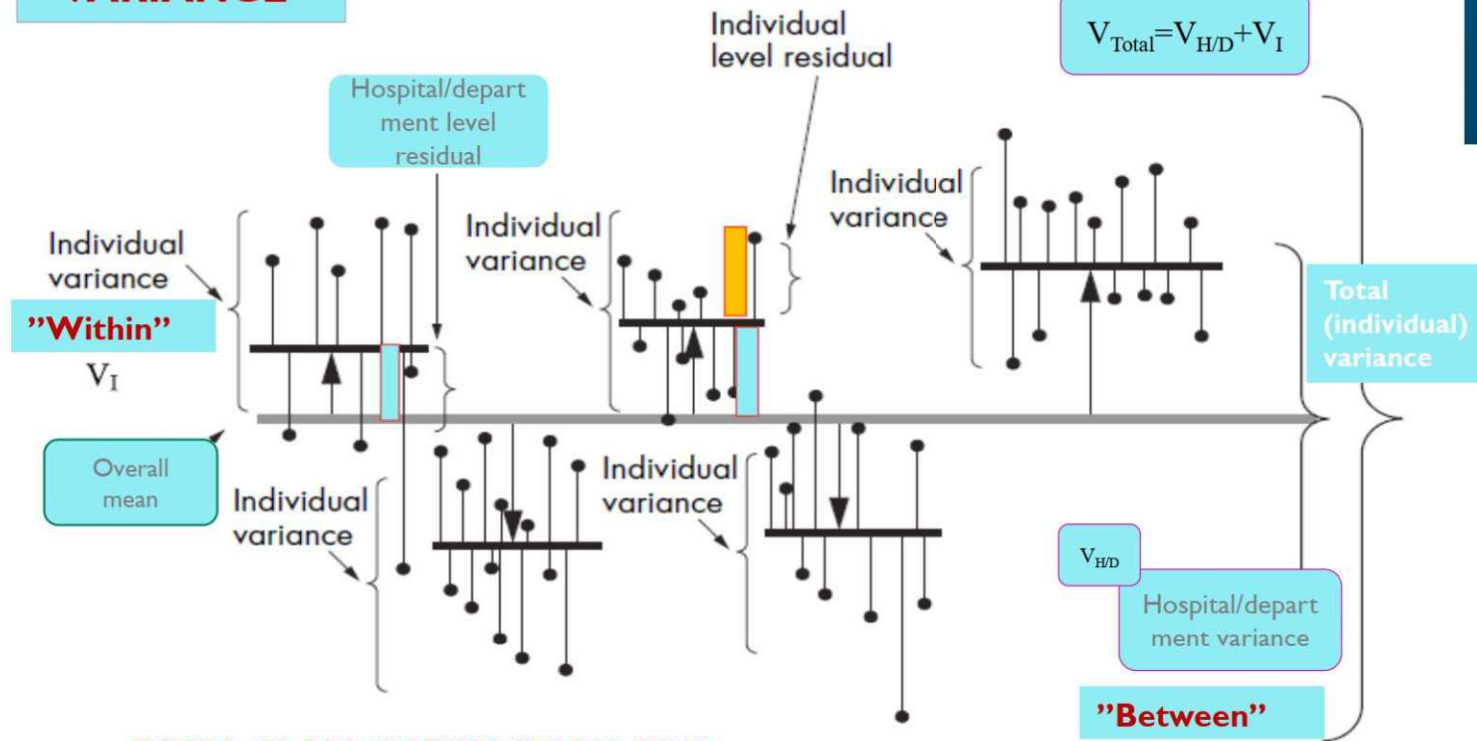
- the **Swedish** patient register with
 - the Cause of Death Register and
 - the Longitudinal Integrated Database for Health Insurance and Labour Studies (LISA).
- **Population:** covered **36,943** patients aged 45-80 who were treated in **565** departments at 71 hospitals.
 - with a discharge diagnosis of heart failure (International Classification of Diseases code I50) ; (2007-2009)



VISUAL STATISTICS...

Multilevel analysis

"VARIANCE"



INTRA-CLASS CORRELATION (ICC)

$$\text{VPC/ICC} = \frac{\text{Between } V_{\text{Hospital/department}}}{\text{between } V_{\text{Hospital/department}} + \text{within } V_I}$$

Total individual variance = within + between



Ghith and Merlo et al (2016): <https://doi.org/10.1371/journal.pone.0148187>



Two step analytical approach

From my PhD Thesis: Variance measures are expressed as median values and 95% credible intervals (CIs)



Patients within departments
within hospitals

	Single level Logistic Regression models		Three Level Multilevel Logistic Regression Models	
	Model 1	Model 2	Model 3	Model 4
Measures of the GCE	(RS) 36 variables on patient case-mix	(RS+gender/age)	(RS+gender/age)	(RS+ gender/age+ patient volume, hospital classification)
AUC	0.727 (0.719-0.736)	0.729 (0.721-0.738)	0.753 (0.745-0.762)	0.752 (0.743-0.760)
ICC _H			0.04% 0.004 (0.000-0.014)	0.002 (0.000-0.015)
ICC _D			5.3% 0.053 (0.035-0.081)	0.036 (0.020-0.064)

Average 30-day mortality was 9 %

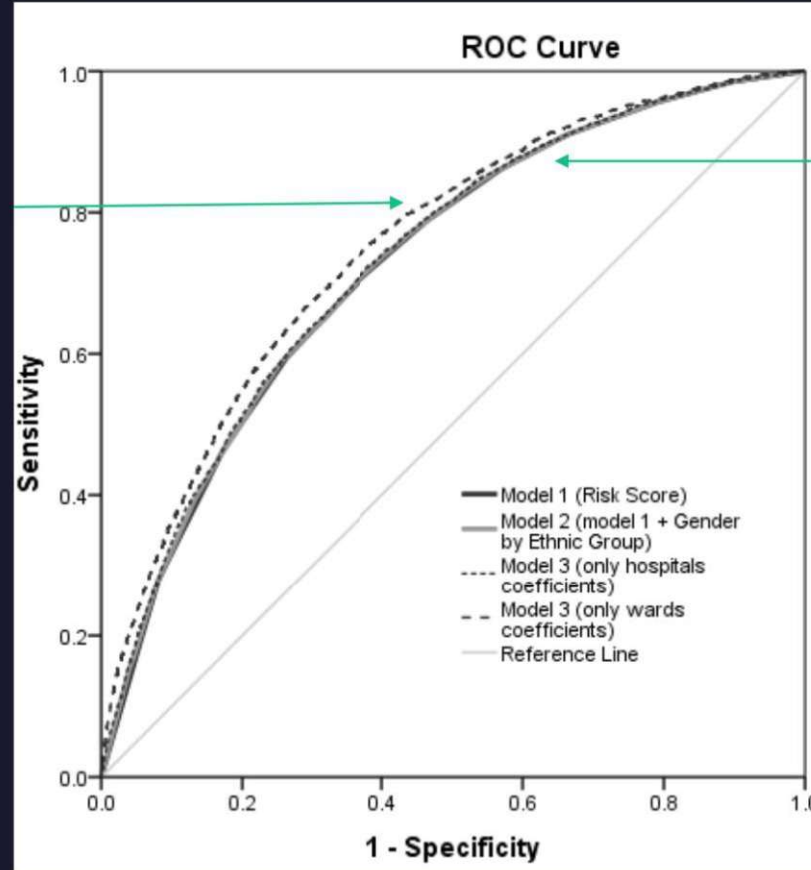


Two step analytical approach

From my PhD Thesis: Variance measures are expressed as median values and 95% credible intervals (CIs)

Multilevel model 3
AUC $\approx$ 0.753

the increase in the AUC in model 3 compared to model 2 represents the *ceiling* of the hospital's general contextual effects.



Single level model 2
AUC = 0.729

Since measures on the GCE were so small, information on the specific contextual effects (patient volume and hospital classification) are not relevant.



RS - quality of patient care and survival

Using the ICC in combination with the average 30-day mortality.

- If the overall survival is good (i.e., high) and the ICC low, all hospitals were doing homogeneously well.
- If the overall survival is low and the ICC low, all hospitals were doing homogeneously bad.
- focus on intervention should be targeting all hospitals in the country.
- However, the higher the ICC, the more appropriate to act on the hospitals with the worst survival.

Using the AUC in combination with the average 30-day mortality.

- If the hospitals and wards do not add discriminatory accuracy, it would not be effective to focus on certain hospitals.
- If we aim to decrease mortality, we should focus on improving care in the patients with higher mortality (using RS) while it does not matter which is the hospital of treatment.

